A Blockchain-Integrated Microfinance & Welfare Management Platform

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

In recent years, digital financial services have become more common, but many users still face difficulties when interacting with these systems. One of the main reasons is the language barrier and the complexity of financial terms used in digital platforms. In countries where people speak multiple languages, users may find it difficult to communicate with systems that support only one language. As a result, many people are unable to easily access information about financial services such as loan eligibility and application procedures. This research focuses on developing a multilingual chatbot using Natural Language Processing (NLP) to improve communication between users and a digital loan advisory system. The chatbot will allow users to ask questions in different languages and receive responses in a simple and understandable way. It will act as an interface that helps users obtain information about loans, eligibility criteria, and related services without needing advanced technical knowledge. The proposed chatbot will use Natural Language Processing techniques to understand user questions and provide relevant responses. It will also connect with other components of the system, such as the loan prediction module and blockchain-based smart contract component, to provide accurate information to users. By doing so, the chatbot will help users interact with the system more easily and efficiently. The system will be developed using technologies such as Python for NLP processing, React.js for the user interface, Node.js for backend services, and NLP libraries such as NLTK or spaCy to handle language understanding. The expected outcome of this research is a simple and user-friendly chatbot that helps users communicate with the system in their preferred language. This can improve accessibility to financial services and make digital platforms easier to use for a wider range of users. In addition, organizations can benefit by providing faster and more efficient automated customer support.

Keywords: Natural Language Processing, Multilingual Chatbot, Financial Services, Conversational System, Artificial Intelligence.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
ML	Machine Learning
API	Application Programming Interface
OCR	Optical Character Recognition
GN	Grama Niladhari
STT	Speech-to-Text

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1. INTRODUCTION

1.1 Background and Context

Globally, public administration has made digital transformation a top priority. In order to provide services like welfare distribution, loan management, and citizen communication, governments are depending more and more on digital platforms. However, local administrative divisions like Grama Niladhari divisions continue to rely largely on manual processes in many developing nations, including Sri Lanka.

When applying for microfinance support or welfare benefits like Samurdhi or Aswesuma, residents often require assistance. Residents and officials frequently communicate via phone calls or in-person visits. Both citizens and administrators must spend a lot of time on this process.

Language diversity presents another difficulty. English, Tamil, and Sinhala are the three most common languages in Sri Lanka. Rural communities find it challenging to use many digital systems because their interfaces are limited to English. New opportunities to enhance citizen interaction with digital services have been brought about by recent advances in artificial intelligence and natural language processing. AI chatbots can automate communication tasks, assist users with administrative processes, and respond to user enquiries.

But traditional chatbots are not without their limitations. They frequently only use pre-trained knowledge and are unable to access domain-specific data, such as welfare policies or government regulations. Inaccurate answers may result from this. This study suggests a multilingual NLP communication module combined with Retrieval Augmented Generation (RAG) to overcome these difficulties. Through the system, citizens will be able to communicate with a digital assistant that can respond in multiple languages and retrieve information from policy documents.

The SmartGrama platform, which aims to enhance digital governance within GN divisions, will incorporate the suggested solution.

1.2 Literature Review

Financial inclusion and access to financial services have been widely discussed in recent research, especially in developing countries where many individuals still face barriers when accessing banking and loan services. According to Saman Kelegama, microfinance institutions play a significant role in supporting low-income communities in Sri Lanka by providing small loans and financial assistance [1]. However, the study highlights that many potential users still struggle to obtain financial information due to communication barriers, lack of awareness, and limited access to digital services. These challenges indicate the need for more accessible digital solutions that can guide users through financial processes.

Several studies have explored the use of conversational systems and Natural Language Processing (NLP) to improve communication between users and digital platforms. A well-known study by Oriol Vinyals and Quoc V. Le introduced a neural conversational model that uses machine learning to generate responses to user queries [2]. This approach demonstrated that conversational systems could automate communication and assist users in retrieving information more efficiently. Later research on chatbot technologies further explored how NLP techniques can be used to recognize user intent, process language input, and generate meaningful responses[9] in conversational interfaces.

In addition, research on multilingual conversational systems shows that language support is an important factor in improving accessibility. Studies such as those by S. S. Gokhale and A. S. Kale proposed multilingual chatbot systems[4] that allow users to interact with digital services in different languages. This approach helps reduce communication barriers and enables users from different linguistic backgrounds to access information more easily. Similarly, researchers from the University of Moratuwa have conducted work on Sinhala and Tamil language processing [11] [13] through their Language Technology Research Laboratory. Their work highlights the importance of developing NLP systems that support local languages, especially in multilingual environments such as Sri Lanka.

Recent advancements in conversational AI frameworks have also contributed to the development of chatbot systems. For example, platforms developed by Rasa Technologies provide open-source tools for building conversational agents [5] using NLP techniques such

as intent recognition and dialogue management. These frameworks allow developers to design chatbot systems that can understand user queries and provide automated responses. Many organizations now use such technologies to develop automated customer support systems and digital assistants.

From a financial services perspective, reports published by the Central Bank of Sri Lanka indicate that financial institutions are increasingly adopting digital technologies [7] such as artificial intelligence, machine learning, and financial technology platforms to improve customer service and operational efficiency. Digital financial services can reduce processing time and improve access to information. However, many existing systems still rely on traditional web interfaces or mobile applications that may be difficult for some users to navigate.

Although these studies and systems demonstrate the potential of NLP and digital financial technologies, several limitations remain. Many chatbot systems focus mainly on English language communication and do not fully support local languages such as Sinhala or Tamil. In addition, some conversational systems provide only basic question-and-answer functions and may not integrate well with financial decision-making systems such as loan evaluation or advisory platforms. Furthermore, users who are unfamiliar with complex financial terminology may still face difficulties when interacting with existing digital platforms.

Based on these limitations, there is a need for a more user-friendly conversational system that supports multilingual communication and provides clear financial guidance. The proposed NLP-based chatbot aims to address these challenges by offering an accessible conversational interface that allows users to interact with a financial advisory system using natural language. By integrating NLP techniques with financial service platforms, the system can improve communication, reduce language barriers, and enhance accessibility for a wider range of users.

1.3 Research Gap

Although previous research has explored chatbot systems and Natural Language Processing (NLP) applications, several important limitations still exist, particularly in the context of financial services in developing countries such as Sri Lanka. Existing chatbot systems, such as those proposed by Vinyals and Le [2] and Gokhale and Kale [4], mainly focus on general conversational tasks or customer service and are primarily designed for English-language interaction. This creates a significant accessibility issue for users who prefer local languages such as Sinhala.

Furthermore, most existing systems do not integrate financial advisory capabilities. Current digital financial platforms typically rely on traditional web or mobile interfaces, requiring users to manually navigate complex forms and financial information. This makes it difficult for individuals with limited financial knowledge or low digital literacy to access essential services.

In addition, although Retrieval-Augmented Generation (RAG) has been introduced in recent studies [6] to improve response accuracy, it has not been widely applied in multilingual financial advisory systems. Most chatbot implementations rely on static or pre-trained knowledge, which can lead to outdated or inaccurate responses when dealing with policy-based or domain-specific information.

Therefore, there is a clear research gap in developing an integrated system that combines multilingual NLP, RAG-based knowledge retrieval, and financial advisory services into a single platform tailored to the Sri Lankan context. The proposed system addresses this gap by providing a user-friendly, multilingual conversational interface capable of delivering accurate, context-aware financial guidance using real-time knowledge sources.

Table 1 Comparison of the existing research methods and proposed method

Technologies and Methods	Research Paper [2]	Research Paper [4]	Research Paper [5]	Research Paper [9]	Research Paper [11]	Proposed System
NLP-based Chatbot	✔	✔	✔	✔	✔	✔
Sinhala English Language Support	✖	✖	✖	✖	✔	✔
Financial / Loan Advisory Domain	✖	✖	✖	✖	✖	✔
Retrieval-Augmented Generation (RAG)	✖	✖	✖	✖	✖	✔
User friendly Conversational Interface	✔	✔	✔	✔	✔	✔
Integration with External Knowledge Sources	✖	✖	✔	✖	✖	✔
Support for Local Context	✖	✖	✖	✖	✖	✔

2 OBJECTIVES

2.1 Main Objective

To design and implement a multilingual AI communication system using Retrieval-Augmented Generation (RAG) that enables citizens to interact with digital welfare and microfinance services through chatbot, speech, and document processing technologies.

2.2 Specific Objectives

1. To develop a multilingual chatbot capable of communicating in Sinhala and English.

    - This objective focuses on developing a chatbot that allows users to interact with the system using both Sinhala and English languages. The chatbot will process user queries written or spoken in either language and generate appropriate responses. Supporting Sinhala is particularly important in the Sri Lankan context because many users prefer communicating in their native language rather than English. By enabling bilingual communication, the system improves accessibility and ensures that a wider range of users can easily obtain information related to financial services and welfare programs.

2. To implement a Retrieval-Augmented Generation framework for retrieving information from policy documents and knowledge bases.

    - The purpose of this objective is to improve the accuracy and reliability of chatbot responses. Instead of generating responses based only on pre-trained knowledge, the system will use a Retrieval-Augmented Generation (RAG) framework. In this approach, the chatbot first retrieves relevant information from a knowledge base containing policy documents, financial guidelines, and frequently asked questions. The retrieved information is then used to generate accurate responses to user queries. This method helps ensure that answers provided by the system are based on verified information sources.

3. To integrate Speech-to-Text functionality for voice-based interaction.

    - This objective focuses on enabling users to interact with the system through voice input**.** The speech-to-text module will convert spoken language into text so that it can be processed by the chatbot. This feature improves usability, especially for users who may find typing difficult or prefer voice interaction. Voice-based communication can also make the system more accessible for elderly users or individuals with limited digital literacy.

4. To develop an OCR module for document processing.

    - This objective aims to implement an Optical Character Recognition (OCR) module that can extract text from uploaded documents such as scanned forms, images, or PDF files. Once the text is extracted, the system can analyze the information and use it to answer user queries or validate document content. This feature reduces the need for manual data entry and helps automate document processing in financial or administrative services.

3. METHODOLOGY

This research follows a Design Science Research (DSR) approach, which focuses on designing and developing a technological artifact to solve a real-world problem. The artifact in this research is a multilingual NLP-based chatbot integrated with a financial advisory system. The design science approach is suitable because the study aims to create and evaluate a functional system that improves communication between users and financial services. The research process includes problem identification, system design, implementation, and evaluation of the developed chatbot.

This research is primarily based on the Artificial Intelligence domain, with a specific focus on Natural Language Processing (NLP) and Information Retrieval techniques. The study also incorporates concepts from Machine Learning and Human-Computer Interaction to enhance user experience and system performance.

The system will be tested using real user queries collected from financial service scenarios in Sri Lanka.
The proposed system uses Natural Language Processing (NLP) techniques to understand user queries and generate relevant responses. Intent recognition methods will be used to identify the purpose of user queries.

To improve response accuracy, the system will incorporate a Retrieval-Augmented Generation (RAG) approach, where the chatbot retrieves relevant information from a predefined knowledge base before generating responses. This method allows the chatbot to provide more accurate and context-aware answers compared to simple rule-based chatbots.

Several tools and technologies will be used to implement the system:

Python – used for implementing NLP algorithms and chatbot logic due to its extensive support for language processing libraries.

NLP Libraries – used for text processing, tokenization, and language understanding.

React.js – used to build the user interface where users interact with the chatbot.

Node.js – used for backend services and API communication between components.

Vector database or document store – used to store knowledge base documents for the RAG retrieval process.

The selection of technologies for this system is based on their suitability and effectiveness in handling NLP and web-based application development.

Python is selected as the primary programming language due to its extensive support for Natural Language Processing through libraries such as NLTK and spaCy. These libraries provide efficient tools for text preprocessing, tokenization, and language understanding.

React.js is chosen for the frontend development because it enables the creation of dynamic and responsive user interfaces, improving user interaction with the chatbot system.

Node.js is used for backend development due to its ability to handle asynchronous operations and real-time communication between system components.

For implementing the Retrieval-Augmented Generation (RAG) approach, a vector database or document store is used to efficiently retrieve relevant information from the knowledge base. This ensures faster and more accurate response generation.

Data may be collected from publicly available financial resources, policy documents, and microfinance information sources. (Sammoopakaraya Bank, Samurdhi Bank)

Ethical considerations will be carefully addressed during the research. Personal user data will not be stored without consent, and any collected data will be used only for research purposes

The performance of the proposed system will be evaluated using both quantitative and qualitative metrics.

Intent recognition accuracy will be measured to determine how accurately the system identifies user intentions from input queries.

Response relevance will be assessed to determine whether the generated responses accurately match user queries and retrieved knowledge.

Response time will be measured to evaluate system efficiency in generating answers within an acceptable time frame.

User satisfaction will be evaluated through user feedback and usability testing, focusing on ease of use, clarity of responses, and overall experience.

These evaluation criteria ensure that the system is not only technically accurate but also user-friendly and effective in real-world scenarios.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed system is a multilingual NLP-based chatbot integrated with a financial advisory platform. The chatbot acts as the main interface between users and the system. Users can ask questions in natural language to obtain information about financial services such as loan eligibility, requirements, and procedures

System Components

- ✓ **User Interface (Frontend)**

  The user interface allows users to interact with the chatbot. It is responsible for collecting user input and displaying responses. The interface can be implemented using React.js and can support web or mobile access.

- ✓ **NLP Processing Module**

  This module processes user input and performs tasks such as text preprocessing, intent recognition, and entity extraction. NLP libraries such as NLTK or spaCy are used to analyze the language and identify the meaning of the user query.

✓ **RAG Retrieval Module**

The Retrieval-Augmented Generation module retrieves relevant information from a knowledge base before generating responses. It improves response accuracy by ensuring that answers are based on verified information sources.

✓ **Backend Service Layer**

The backend service manages communication between system components. It processes requests, connects to databases, and sends results back to the chatbot interface. Technologies such as Node.js or Python-based frameworks can be used for backend development.

✓ **Knowledge Base / Data Source**

The knowledge base stores financial information, loan guidelines, and frequently asked questions. The RAG module retrieves relevant documents from this database to generate responses.

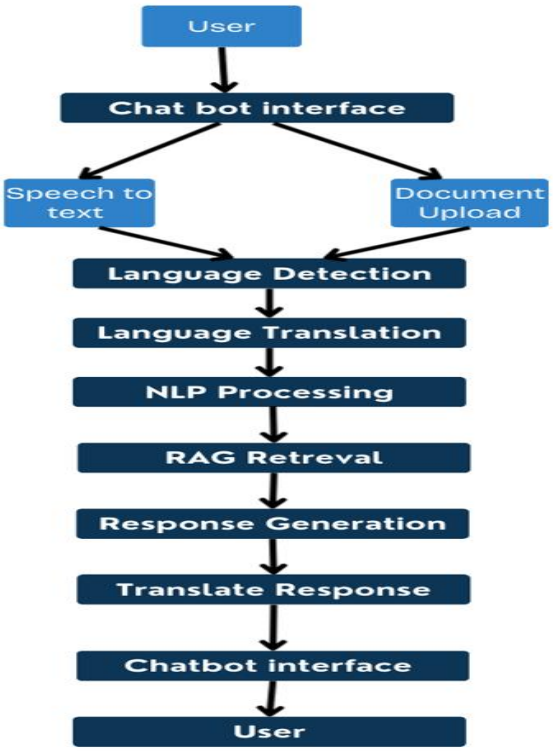


Figure 1 System architecture

The data flow in the system begins when a user submits a query or uploads a document. The input is processed through the NLP module, which identifies the intent of the query. The RAG retrieval system then retrieves relevant information from the knowledge base to generate an accurate response. The final response is returned to the chatbot interface, where it is displayed to the user.

Security measures include secure data transmission, access control mechanisms, and data privacy protection. Sensitive user information and uploaded documents will be handled securely to prevent unauthorized access.

5. USER REQUIREMENTS

Several stakeholders interact with or benefit from the system. The primary stakeholders are residents within the Grama Niladhari (GN) division, who will use the chatbot to obtain information about welfare programs, microfinance services, and administrative procedures. Grama Niladhari officers and Samurdhi officers are also important stakeholders since the system can help reduce their communication workload by automatically answering common questions from citizens. In addition, microfinance institutions may use the platform to provide loan-related guidance and interact with residents. Finally, system administrators and government authorities are responsible for maintaining the system and ensuring that the information stored in the knowledge base remains accurate and up to date.

Functional Requirements

1. The system shall allow users to interact with the platform through a chatbot interface.
2. The system shall support communication in Sinhala and English languages.
3. The system shall allow users to submit queries through both text and voice input.
4. The system shall convert voice input into text using a speech-to-text module.
5. The system shall retrieve relevant information from policy documents using the RAG architecture.
6. The system shall generate accurate responses based on retrieved information.
7. The system shall allow users to upload documents such as PDF forms or images.
8. The system shall extract textual information from uploaded documents using OCR technology.

Non-Functional Requirements

Security

The system must ensure that user data and uploaded documents are securely stored and transmitted. Authentication and secure communication protocols should be implemented to prevent unauthorized access.

Performance

The system should process user queries and generate responses within a few seconds to provide a smooth user experience.

Usability

The interface should be simple and easy to use, especially for users with limited technical knowledge. Clear instructions and multilingual support should be provided.

Scalability

The system should be capable of handling multiple users simultaneously without performance degradation.

Reliability

The system should provide accurate responses and maintain stable operation during continuous use.

Use Case Example

A user interacts with the chatbot by asking a question such as “Am I eligible for a loan?” in Sinhala or English. The system processes the query using the NLP module to identify the user’s intent. The RAG module retrieves relevant information from the knowledge base, such as loan eligibility criteria. The chatbot then generates a clear and simple response, providing guidance to the user. This interaction demonstrates how the system simplifies access to financial information.

6. COMMERCIALIZATION PLAN

The primary target market for the proposed system includes financial institutions, microfinance organizations, and digital banking platforms that require automated customer interaction systems. These organizations often handle large numbers of customer inquiries related to loans, financial services, and eligibility criteria.

Another important target group is individual users who seek financial guidance, particularly people with limited access to financial information or those who prefer communicating in their native language. This includes small business owners, rural communities, and individuals who rely on microfinance services.

The system can generate revenue through several models:

Software-as-a-Service (SaaS) Model

Financial institutions can subscribe to the chatbot platform and pay a monthly or annual fee to use the service.

Customization and Integration Services

Organizations may require customized versions of the chatbot integrated with their existing systems.

Licensing Model

The chatbot platform can be licensed to financial institutions or microfinance organizations for long-term use.

The main costs involved in developing and deploying the system include:

- **Development costs:** software development, system design, and testing
- **Infrastructure costs:** cloud hosting
- **Maintenance costs:** system updates, bug fixing, and performance monitoring
- **Operational costs:** technical support and system administration

The proposed system offers several advantages compared to existing chatbot systems:

- Integration of Natural Language Processing with financial advisory systems
- Support for multiple languages, which improves accessibility
- Ability to provide financial guidance in a conversational format
- Improved user experience compared to traditional web-based financial platforms

These features make the system particularly useful for organizations serving diverse user populations.

The increasing adoption of digital financial services in Sri Lanka highlights a strong market demand for intelligent and user-friendly systems. Many financial institutions are actively investing in digital transformation to improve customer experience and operational efficiency. The proposed system aligns with this trend by providing an AI-driven solution that enhances accessibility and communication.

By targeting microfinance institutions, government agencies, and digital banking platforms, the system has strong potential for real-world implementation. The ability to support multilingual communication and provide financial guidance makes it especially valuable for rural and underserved communities.

7. BUDGET AND JUSTIFICATION

The development of the proposed system requires several operational resources such as internet access, electricity, transportation, and communication. These resources support system development, testing, research activities, and collaboration with supervisors and team members.

Table 2 Budget Estimated

Resources	Cost (Rs)
Internet	4,000
Electricity	5,000
Transport	7,000
Communication	2,000
Purchasing	10,000
Paper Publish Cost	5,000
Total Estimated Budget	33,000

Internet (LKR 4,000)

Internet access is essential for conducting literature reviews, accessing online research databases, downloading blockchain development tools, and maintaining communication with team members and supervisors.

Electricity (LKR 5,000)

Electricity is required for operating computers, running blockchain development environments, testing system components, and maintaining continuous development work.

Transport (LKR 7,000)

Transportation expenses are associated with traveling for project meetings, research discussions, and consultations with supervisors or team members.

Communication (LKR 2,000)

Communication costs include phone calls, online meetings, and collaboration with team members during the research and development process.

Purchasing (LKR 10,000)

This budget covers purchasing essential development resources such as hardware accessories, development tools, or software resources required during the project.

Paper Publish Cost (LKR 5,000)

This cost is allocated for publishing the research paper in academic conferences or journals to present the research findings and contribute to academic knowledge.

8.WORK BREAKDOWN STRUCTURE (WBS)

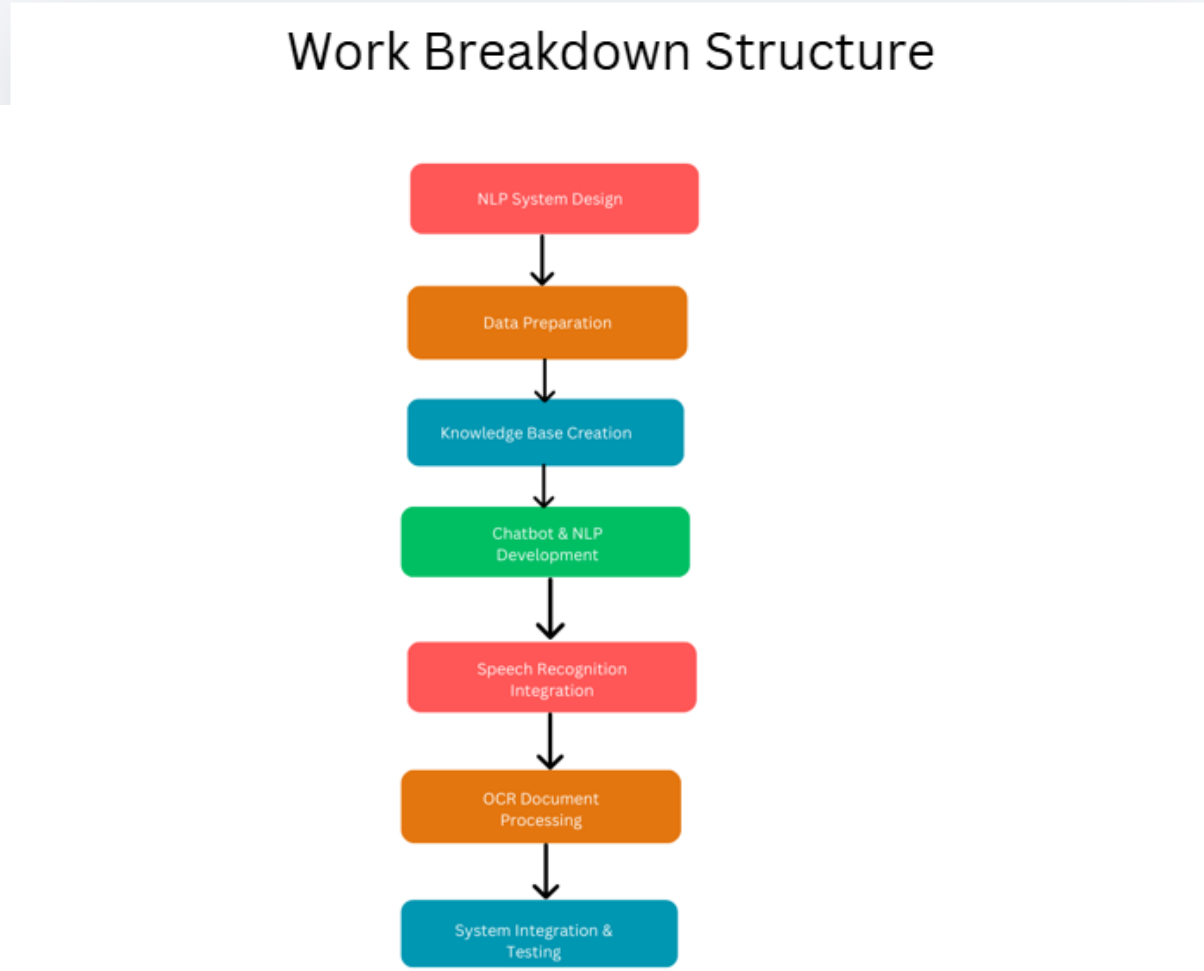


Figure 2 work break down structure

The Work Breakdown Structure (WBS) outlines the major phases of the project, including requirement analysis, system design, development, testing, and deployment. Each phase is divided into smaller tasks to ensure effective planning and workload distribution. Time estimates are allocated for each activity to ensure that the project is completed within the given timeline. This structured approach helps manage resources efficiently and track project progress.

9. GANTT CHART

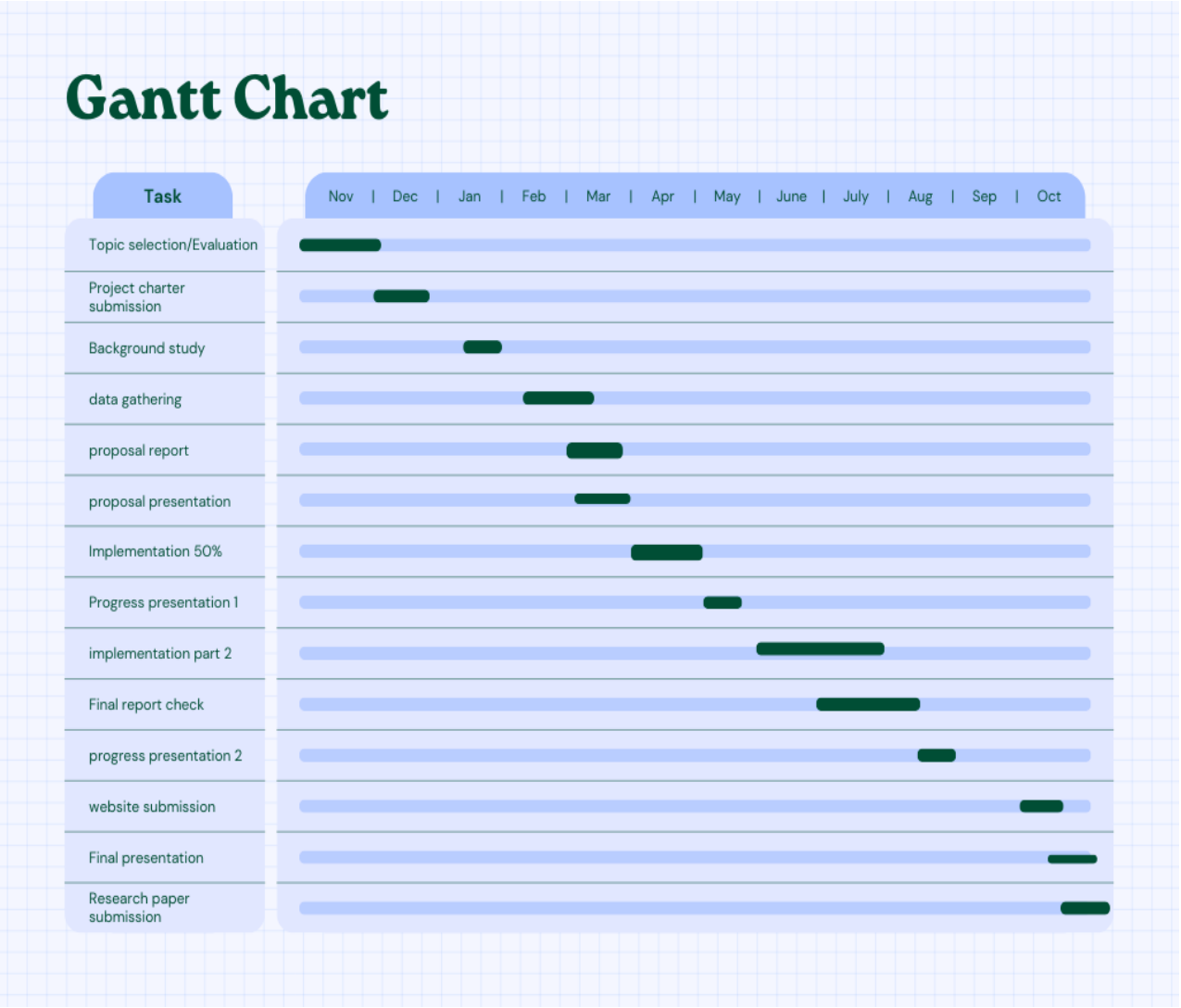


Figure 3 Gantt chart

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11. APPENDICES

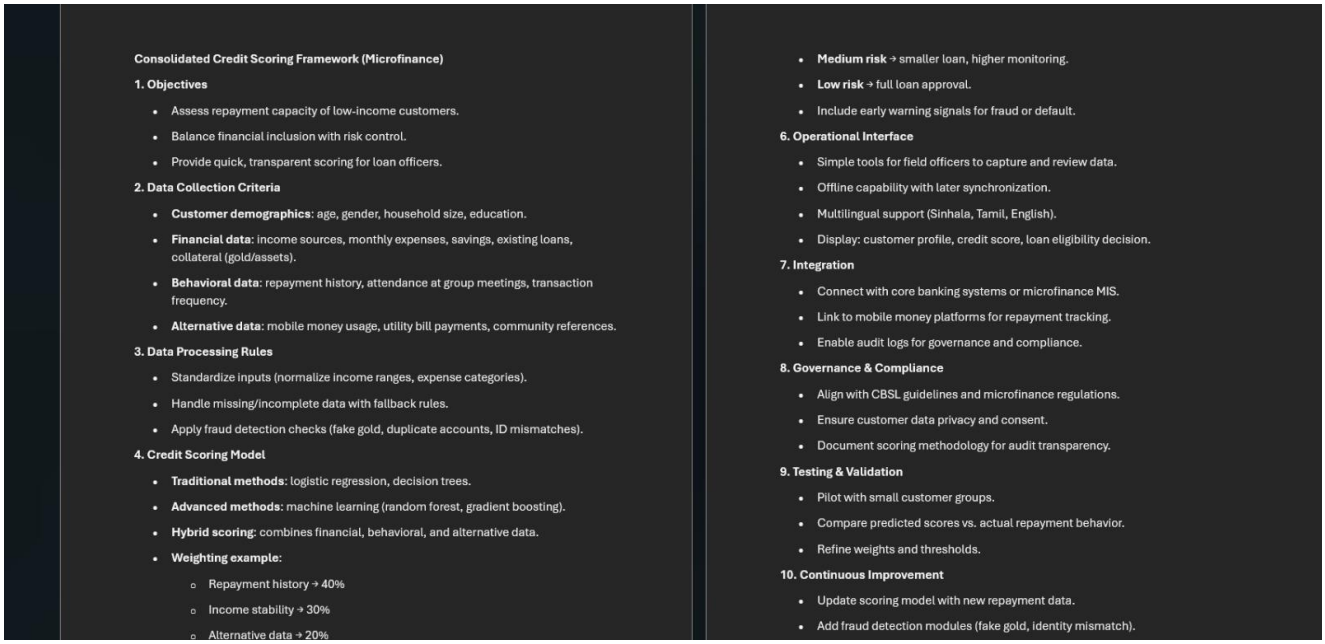
Sample Ledger Portfolio (Microfinance Credit Scoring)

Field	Details / Example	Purpose
Customer ID / Account No.	2025-MF-001	Unique identifier
Customer Name	John Silva	Identification
National ID / Passport No.	123456789V	Verification
Age / Gender	35 / Male	Demographics
Address / Location	Colombo, Sri Lanka	Regional mapping
Household Size	4	Dependency ratio
Loan Type	Gold-backed / Micro loan	Product category
Loan Amount	LKR 50,000	Exposure
Collateral Details	22K Gold chain, 25g	Security
Monthly Income	LKR 35,000	Repayment capacity
Monthly Expenses	LKR 20,000	Debt-to-income ratio
Savings / Deposits	LKR 5,000	Financial discipline
Existing Loans	LKR 15,000 (other MFI)	Over-indebtedness check
Repayment History	95% on-time	Reliability
Attendance (Group Meetings)	90%	Commitment
Mobile Money Usage	Active (weekly transactions)	Alternative data
Utility Bill Payments	On-time	Responsibility
Risk Flags	None	Fraud detection
Credit Score	78/100 (Band B)	Risk rating
Decision	Approved – Conditional	Loan eligibility
Officer Remarks	Customer reliable, collateral verified	Governance
Date of Entry	08 Dec 2025	Audit trail

✅ **Key Notes**

- This **ledger portfolio** can be digitized into your app database.
- Each row = one customer record.
- You can add **columns for audit logs** (who entered, who approved).

Appendix 1 Loan Application Data Attributes (Provided by Samupakara Gramiya Sanwardhana Bank)



Consolidated Credit Scoring Framework (Microfinance)

1. Objectives

- Assess repayment capacity of low-income customers.
- Balance financial inclusion with risk control.
- Provide quick, transparent scoring for loan officers.

2. Data Collection Criteria

- **Customer demographics**: age, gender, household size, education.
- **Financial data**: income sources, monthly expenses, savings, existing loans, collateral (gold/assets).
- **Behavioral data**: repayment history, attendance at group meetings, transaction frequency.
- **Alternative data**: mobile money usage, utility bill payments, community references.

3. Data Processing Rules

- Standardize inputs (normalize income ranges, expense categories).
- Handle missing/incomplete data with fallback rules.
- Apply fraud detection checks (fake gold, duplicate accounts, ID mismatches).

4. Credit Scoring Model

- **Traditional methods**: logistic regression, decision trees.
- **Advanced methods**: machine learning (random forest, gradient boosting).
- **Hybrid scoring**: combines financial, behavioral, and alternative data.
- **Weighting example**:
  - Repayment history → 40%
  - Income stability → 30%
  - Alternative data → 20%
- **Medium risk** → smaller loan, higher monitoring.
- **Low risk** → full loan approval.
- Include early warning signals for fraud or default.

6. Operational Interface

- Simple tools for field officers to capture and review data.
- Offline capability with later synchronization.
- Multilingual support (Sinhala, Tamil, English).
- Display: customer profile, credit score, loan eligibility decision.

7. Integration

- Connect with core banking systems or microfinance MIS.
- Link to mobile money platforms for repayment tracking.
- Enable audit logs for governance and compliance.

8. Governance & Compliance

- Align with CBSL guidelines and microfinance regulations.
- Ensure customer data privacy and consent.
- Document scoring methodology for audit transparency.

9. Testing & Validation

- Pilot with small customer groups.
- Compare predicted scores vs. actual repayment behavior.
- Refine weights and thresholds.

10. Continuous Improvement

- Update scoring model with new repayment data.
- Add fraud detection modules (fake gold, identity mismatch).

Appendix 2 Bank Requirement Document